Q. Show that A is diagonalizable and diagonalize A.

$$A=\begin{pmatrix}4 & 1 & -1\\ 2 & 5 & -2\\ 1 & 1 & 2\end{pmatrix}$$

Soln: $|A-\lambda I|=0$

$$\Rightarrow \begin{vmatrix}4-\lambda & 1 & -1\\ 2 & 5-\lambda & -2\\ 1 & 1 & 2-\lambda\end{vmatrix}=0$$

$$\Rightarrow (4-\lambda)[(5-\lambda)(2-\lambda)+2]-1[2(2-\lambda)+2]-1[2-5+\lambda]=0$$

$$\Rightarrow (4-\lambda)[10-7\lambda+\lambda^2]-(6-2\lambda)-(\lambda-3)=0$$

$$\Rightarrow 48-28\lambda+4\lambda^2-12\lambda+7\lambda^2-\lambda^3-6+2\lambda-\lambda+3=0$$

$$\Rightarrow \lambda^3+39\lambda-11\lambda^2-45=0$$

$\therefore \lambda = 5, 3, 3.$

For $\lambda=5$, let $\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}$ be the eigenvector

$$\begin{pmatrix}4-\lambda & 1 & -1\\ 2 & 5-\lambda & -2\\ 1 & 1 & 2-\lambda\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-1 & 1 & -1\\ 2 & 0 & -2\\ 1 & 1 & -3\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\left(\begin{array}{ccc|c}-1 & 1 & -1 & 0\\ 2 & 0 & -2 & 0\\ 1 & 1 & -3 & 0\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}-1 & 1 & -1 & 0\\ 2 & 0 & -2 & 0\\ 1 & 1 & -3 & 0\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}-1 & 1 & -1 & 0\\ 0 & 2 & -4 & 0\\ 0 & 2 & -4 & 0\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}-1 & 1 & -1 & 0\\ 0 & 2 & -4 & 0\\ 0 & 0 & 0 & 0\end{array}\right)$$

$-x_1+x_2-x_3=0$

$2x_2-4x_3=0$

Let, $x_3=t$ [free variable]

$\therefore 2x_2=4x_3 \Rightarrow x_2=2t$

$\therefore x_1=t$

$$\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}t\\ 2t\\ t\end{pmatrix}=t\begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$$

$$V_1=\begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$$

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For $\lambda=3$, let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector.

$$\begin{pmatrix} 4-\lambda & 1 & -1 \\ 2 & 5-\lambda & -2 \\ 1 & 1 & 2-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\therefore x_1+x_2-x_3=0$

Let, $x_2=t_1$ [free variable]

$x_3=t_2$

$x_1=-t_1+t_2$

$$\sim\left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 2 & 2 & -2 & 0 \\ 1 & 1 & -1 & 0 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 2 & 2 & -2 & 0 \\ 1 & 1 & -1 & 0 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} -t_1+t_2 \\ t_1 \\ t_2 \end{pmatrix}=\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}t_1+\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}t_2$$

$V_2=\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ and $V_3=\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$\lambda=5$	$\lambda=3$	$\lambda=3$
$V_1=\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$	$V_2=\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$	$V_3=\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$$D=\begin{pmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \qquad P=\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$P^{-1}=\begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$

$PDP^{-1}=\ldots\ldots$

$=\ldots\ldots$

$=\ldots\ldots$

$=$

$PDP^{-1}=A$ [Verify]

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9. Find the eigenvectors of A where $-3, -2, -2$ are the eigenvalues of the matrix A.

$$A = \begin{pmatrix} 0 & -6 & -4 \\ 5 & -11 & -6 \\ -6 & 9 & 4 \end{pmatrix}$$

Sol^n: $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 0-\lambda & -6 & -4 \\ 5 & -11-\lambda & -6 \\ -6 & 9 & 4 \end{vmatrix} = 0$$

$\Rightarrow \cdots\cdots\cdots$

$\therefore \lambda = -3, -2, -2$

For $\lambda = -3$: let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector

$$\begin{pmatrix} 0-\lambda & -6 & -4 \\ 5 & -11-\lambda & -6 \\ -6 & 9 & 4-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 3 & -6 & -4 \\ 5 & -8 & -6 \\ -6 & 9 & 7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$3x_1 - 6x_2 - 4x_3 = 0$

$6x_2 + 2x_3 = 0$

Let, $x_3 = t$ [free]

$\therefore x_2 = -\frac{t}{3}$

$$x_1 = \frac{-2t + 4t}{3} = \frac{+6t}{3} = +2t$$

$$\begin{pmatrix} 3 & -6 & -4 & 0 \\ 5 & -8 & -6 & 0 \\ -6 & 9 & 7 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 3 & -6 & -4 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & -3 & -1 & 0 \end{pmatrix}$$

$$\sim \begin{pmatrix} 3 & -6 & -4 & 0 \\ 0 & 6 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} +2t \\ -\frac{t}{3} \\ t \end{pmatrix} = t \begin{pmatrix} +2 \\ -\frac{1}{3} \\ 1 \end{pmatrix} = \frac{t}{3} \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

$$V_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

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For $\lambda = -2$, let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector:

$$\begin{pmatrix} 0-\lambda & -6 & -4 \\ 5 & -11-\lambda & -6 \\ -6 & 9 & 4-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2 & -6 & -4 \\ 5 & -9 & -6 \\ -6 & 9 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$2x_1 - 6x_2 - 4x_3 = 0$$
$$12x_2 + 8x_3 = 0$$

$x_3 = t$ [free variable]

$x_2 = \frac{-2t}{3}$

~~$x_1 = \frac{2t}{1} + \frac{2t}{1} - \frac{4t}{1}$~~

$x_1 = 3x_2 + 2x_3 = -2t + 2t = 0$

$$\sim \left(\begin{array}{ccc|c} 2 & -6 & -4 & 0 \\ 5 & -9 & -6 & 0 \\ -6 & 9 & 6 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -6 & -4 & 0 \\ 0 & 12 & 8 & 0 \\ 0 & -9 & -6 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -6 & -4 & 0 \\ 0 & 12 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -2t/3 \\ t \end{pmatrix} = t/3 \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$$

$$V_2 = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$$

$\lambda = -3$	$\lambda = -2, -2$
$V_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$	$V_2 = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$

$$D = \begin{pmatrix} -3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad P = \begin{pmatrix} 2 & 0 & 0 \\ -1 & -2 & -2 \\ 3 & 3 & 3 \end{pmatrix}$$

P^{-1} does not exist

Not A diagonalizable matrix

As P is not invertible

$PDP^{-1} \neq A$.

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9. Find the value of A^{100} where $A=\begin{pmatrix}6 & -1\\ 2 & 3\end{pmatrix}$.

Soln: $|A-\lambda I|=0$

$$\begin{vmatrix}6-\lambda & -1\\ 2 & 3-\lambda\end{vmatrix}=0$$

$\Rightarrow (6-\lambda)(3-\lambda)-(-1)2=0$

$\Rightarrow 18-9\lambda+\lambda^2+2=0$

$\Rightarrow \lambda^2-9\lambda+20=0$

$\therefore \lambda=5,4.$

For $\lambda=5$, let $\begin{pmatrix}x_1\\ x_2\end{pmatrix}$ be the eigenvector

$$\begin{pmatrix}6-\lambda & -1\\ 2 & 3-\lambda\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}$$

$\Rightarrow \begin{pmatrix}1 & -1\\ 2 & -2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}$

$\sim \left(\begin{array}{cc|c}1 & -1 & 0\\ 2 & -2 & 0\end{array}\right)$

$\sim \left(\begin{array}{cc|c}1 & -1 & 0\\ 0 & 0 & 0\end{array}\right)$

$x_1-x_2=0$

let, $x_2=t$ [free variable]

$\therefore x_1=t$

$\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}t\\ t\end{pmatrix}=t\begin{pmatrix}1\\ 1\end{pmatrix}$

$\therefore V_1=\begin{pmatrix}1\\ 1\end{pmatrix}t$

For $\lambda=4$, let $\begin{pmatrix}x_1\\ x_2\end{pmatrix}$ be the eigenvector

$$\begin{pmatrix}6-\lambda & -1\\ 2 & 3-\lambda\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}$$

$\Rightarrow \begin{pmatrix}2 & -1\\ 2 & -1\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}$

$\sim \left(\begin{array}{cc|c}2 & -1 & 0\\ 2 & -1 & 0\end{array}\right)$

$\sim \left(\begin{array}{cc|c}2 & -1 & 0\\ 0 & 0 & 0\end{array}\right)$

$2x_1-x_2=0$

$x_2=t$ [free variable]

$\therefore x_1=t/2$

$\begin{pmatrix}x_1\\ x_2\end{pmatrix}=\begin{pmatrix}t/2\\ t\end{pmatrix}=\begin{pmatrix}1/2\\ 1\end{pmatrix}t=t/2\begin{pmatrix}1\\ 2\end{pmatrix}$

$V_2=\begin{pmatrix}1\\ 2\end{pmatrix}$

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$\lambda_1 = 5$ | $\lambda_2 = 4$

$V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ | $V_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$

$P^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$

$D = \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}$

$P^{-1}AP = D$

$A = PDP^{-1}$

$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$

$= \begin{pmatrix} 5 & 4 \\ 5 & 8 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$

$A = \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}$

$D = \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}$

$D^2 = \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 25 & 0 \\ 0 & 16 \end{pmatrix} = \begin{pmatrix} 5^2 & 0 \\ 0 & 4^2 \end{pmatrix}$

$D^3 = \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 125 & 0 \\ 0 & 64 \end{pmatrix} = \begin{pmatrix} 5^3 & 0 \\ 0 & 4^3 \end{pmatrix}$

$D^{100} = \begin{pmatrix} 5^{100} & 0 \\ 0 & 4^{100} \end{pmatrix}$.

Proof: $A = PDP^{-1}$

$A^2 = A \cdot A = PDP^{-1} \cdot PDP^{-1} = PDIDP^{-1} = PD^2P^{-1}$

$A^3 = A^2 \cdot A = PD^2P^{-1} \cdot PDP^{-1} = PD^2IDP^{-1} = PD^3P^{-1}$

$A^{100} = PD^{100}P^{-1}$.

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$$A = PDP^{-1}$$

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$A^{100} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 4 \end{pmatrix}^{100} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5^{100} & 0 \\ 0 & 4^{100} \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5^{100} & 4^{100} \\ 5^{100} & 2.4^{100} \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2.5^{100} - 4^{100} & -5^{100} + 4^{100} \\ 2.5^{100} - 2.4^{100} & -5^{100} + 2.4^{100} \end{pmatrix}$$

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Fourier Series

*Periodic Functions:

A function $f(x)$ is said to have a period P or to be periodic with period P if for all x, $\boxed{f(x+P) = f(x)}$:

The least value of $P > 0$ is called the least period or simply the period of $f(x)$.

Example: The function $\sin x$ has periods $2\pi, 4\pi, 6\pi, \ldots\ldots$, since $\sin(x+2\pi)$, $\sin(x+4\pi)$, $\sin(x+6\pi)$ .... all equal $\sin(x)$. However 2π is the least period or period of $\sin(x)$.

Period of Trigonometric Functions:

$\sin(x), \cos(x), \sec(x), \operatorname{cosec}(x) \longrightarrow$ Period : 2π.

$\tan(x), \cot(x) \longrightarrow$ Period : π.

If a given function is in the form below:

$$f(x) = A \sin^b(cx+d) + e$$

then, $\text{Period} = \dfrac{\text{Main Period}}{c}$

$\sin(x), \cos(x), \sec(x), \operatorname{cosec}(x) \rightarrow$ Period : 2π

$\sin^2(x), \cos^2(x), \sec^2(x), \operatorname{cosec}^2(x) \rightarrow$ Period : π

$\sin^3(x), \cos^3(x), \sec^3(x), \operatorname{cosec}^3(x) \rightarrow$ Period : 2π

For Even Power : Period $= \pi$

For Odd Power : Period $= 2\pi$.

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Q.1. Determine the period of the following function

$$f(x) = 1952 \cos^{1971}(2022x + 2023)$$

Soln: For $\cos(x) \rightarrow$ Period $= 2\pi$

For $\cos^{1971}(x) \rightarrow$ Period $= 2\pi$

For $\cos^{1971}(2022x) \rightarrow$ Period $= \frac{2\pi}{2022}$

$\therefore$ For $1952 \cos^{1971}(2022x + 2023) \rightarrow$ Period $= \frac{2\pi}{2022}$

Q.2. Determine the period of the function

$y = 2023 \sin^{2022}(11x + 12)$.

Soln: For $\sin(x) \rightarrow$ Period $= 2\pi$

$\sin^{2022}(x) \rightarrow$ Period $= \pi$

$\sin^{2022}(11x) \rightarrow$ Period $= \frac{\pi}{11}$

$2023 \sin^{2022}(11x + 12) \rightarrow$ Period $= \frac{\pi}{11}$

Q.3. Determine the period of the function

$$f(x) = \tan^{2022}(2022x + 2023)$$

Soln: For $\tan(x) \rightarrow$ Period $= \pi$

$\tan^{2022}(x) \rightarrow$ Period $= \pi$

$\tan^{2022}(2022x) \rightarrow$ Period $= \frac{\pi}{2022}$

$\tan^{2022}(2022x + 2023) \rightarrow$ Period $= \frac{\pi}{2022}$.

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Q. Draw sketch of the following functions;

(a) $S(x) = \frac{x}{\pi}$ for $-\pi \leq x < \pi$

(b) $S(x) = 3|\sin x|$ for $0 \leq x < 2\pi$

(c) $S(x) = \begin{cases} 2\sin x & \text{for } 0 \leq x < \pi \\ 0 & \text{for } \pi \leq x < 2\pi \end{cases}$

(d) $S(x) = \begin{cases} 1 & \text{for } 0 \leq x < \pi \\ 0 & \text{for } \pi \leq x < 2\pi \end{cases}$

(e) $S(x) = A - \frac{Ax}{P}$ for $0 \leq x < P$

(f) $f(x) = \begin{cases} x & \text{for } 0 < x < \pi \\ -x & \text{for } -\pi < x \leq 0 \end{cases}$

(g) $f(x) = \begin{cases} 1 & \text{when } |x| \leq 1 \\ 0 & \text{otherwise.} \end{cases}$

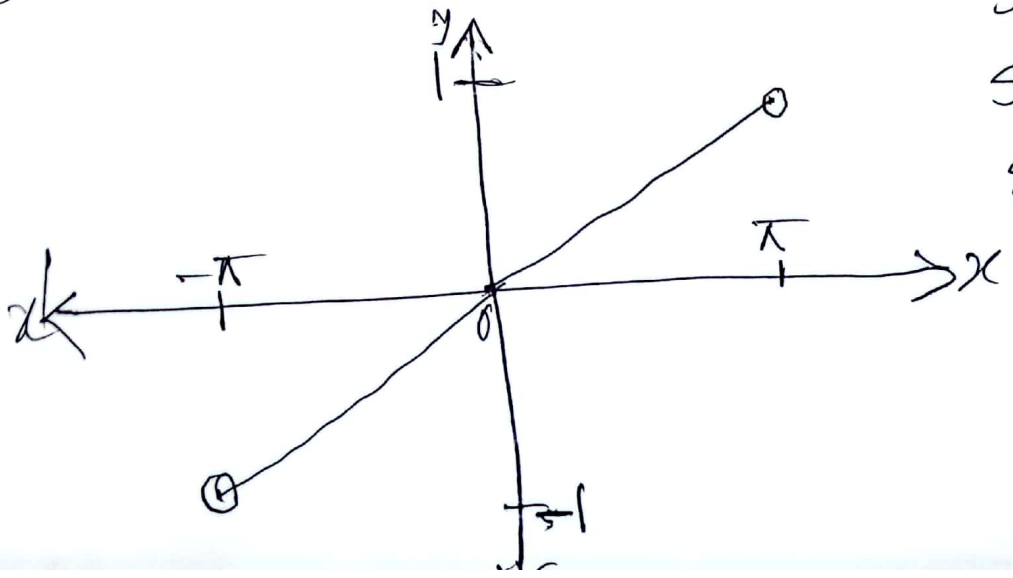
(h) $f(x) = \begin{cases} 1 & \text{when } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$

(a) Soln: $S(x) = \frac{x}{\pi}$ $[-\pi \leq x < \pi]$

$S(0) = \frac{0}{\pi} = 0$

$S(\pi) = \frac{\pi}{\pi} = 1$

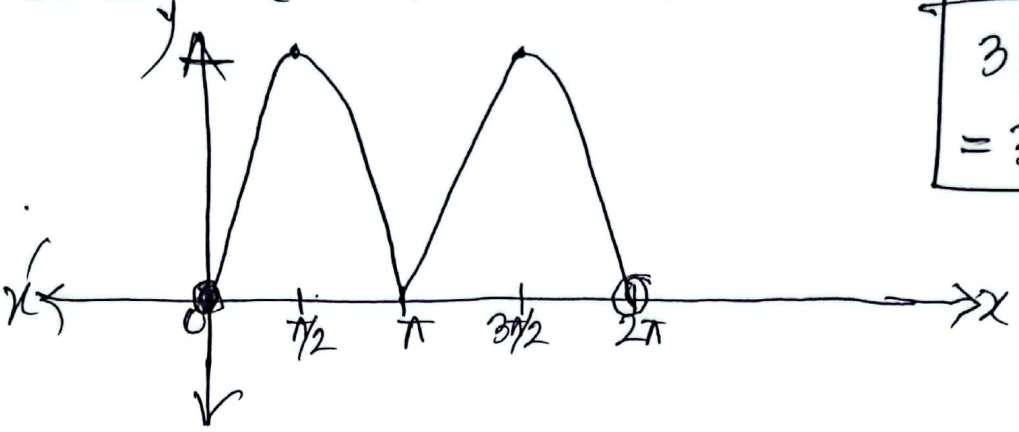
$S(-\pi) = \frac{-\pi}{\pi} = -1$



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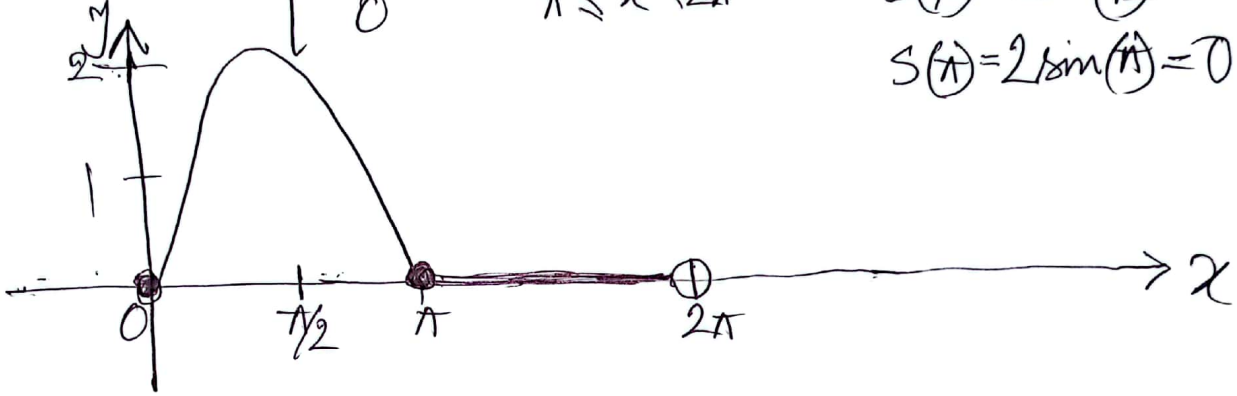
(b) Soln. $S(x) = 3|\sin(x)|$ $[0 \le x < 2\pi]$

$3|\sin 3\pi/2|$
$= 3(1) = 3$



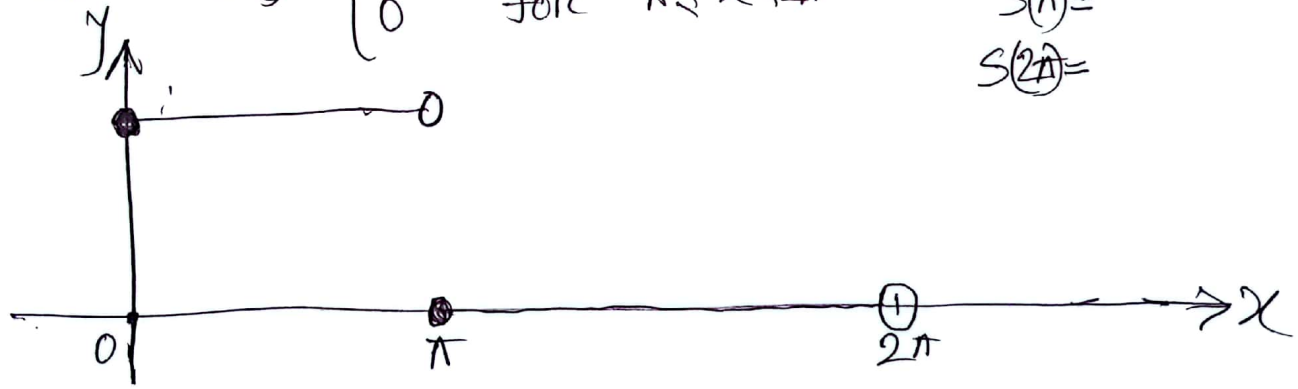
(c) Soln. $S(x) = \begin{cases} 2\sin(x) & \text{for } 0 \le x < \pi \\ 0 & \pi \le x < 2\pi \end{cases}$

$S(0) = 2\sin(0) = 0$
$S(\pi/2) = 2\sin(\pi/2) = 2$
$S(\pi) = 2\sin(\pi) = 0$

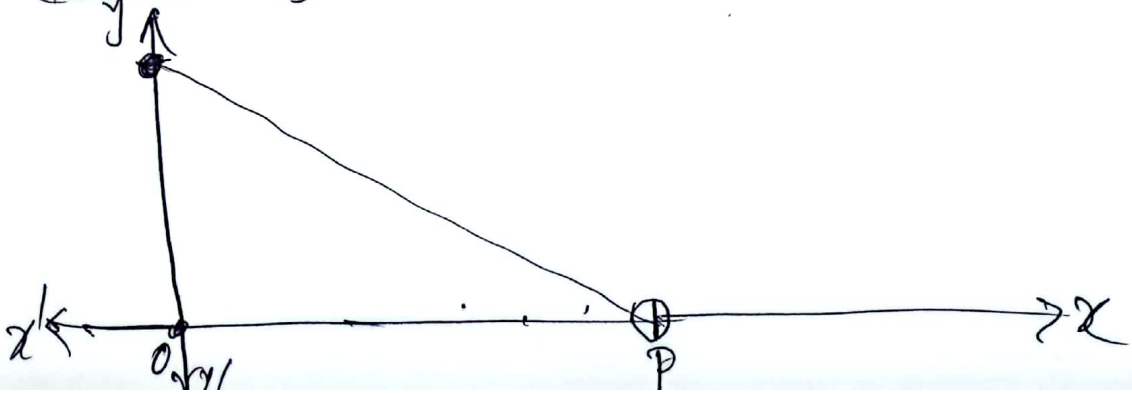


(d) Soln. $S(x) = \begin{cases} 1 & \text{for } 0 \le x < \pi \\ 0 & \text{for } \pi \le x < 2\pi \end{cases}$

$S(0) =$
$S(\pi) =$
$S(2\pi) =$



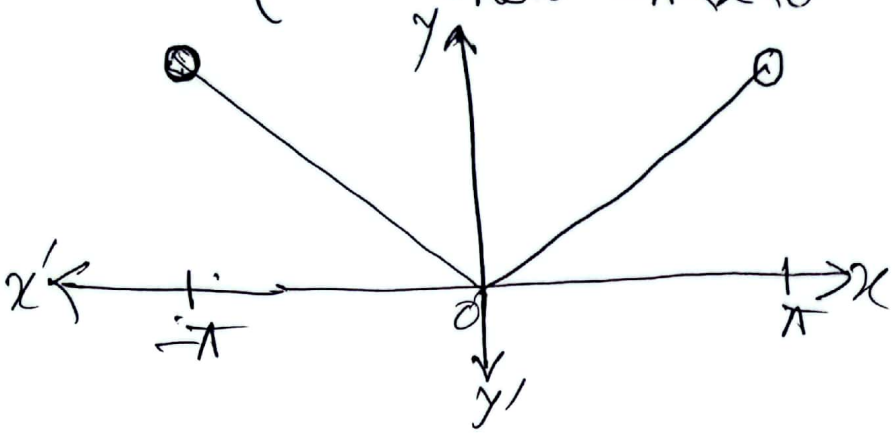
(e) Soln. $S(x) = A - \frac{A}{P}x$ for $0 \le x < P$



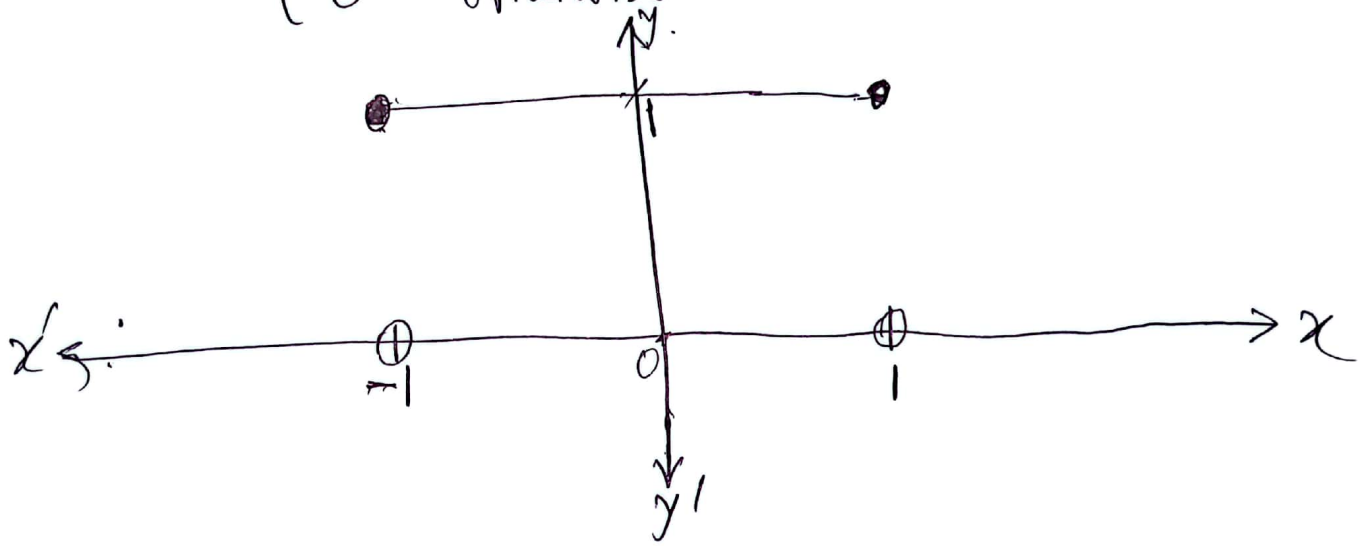
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(f) $f(x) = \begin{cases} x & \text{when } 0 < x < \pi \\ -x & \text{when } -\pi < x < 0 \end{cases}$

$f(-\pi) = -(-\pi) = \pi$
$f(0) = 0$
$f(\pi) = \pi$



(g/h) $f(x) = \begin{cases} 1 & \text{when } (|x| \le 1) \text{ or } (-1 \le x \le 1). \\ 0 & \text{otherwise} \end{cases}$



* Odd / Even Functions:

$f(-x) = f(x) \rightarrow$ Even

$f(-x) = -f(x) \rightarrow$ Odd

Even: $f(-x) = f(x)$

$f(x) = x^4$

$f(x) = \cos(x)$

$f(x) = x \sin(x)$

$f(-x) = (-x)^4 = x^4 = f(x)$

$f(-x) = \cos(-x) = \cos x = f(x)$

$f(-x) = (-x)\sin(-x) = x \sin x = f(x)$

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Odd: $f(-x) = -f(x)$

$f(x) = x^3$ $\quad f(-x) = (-x)^3 = -x^3 = -f(x)$

$f(x) = \sin(x)$ $\quad f(-x) = \sin(-x) = -\sin x = -f(x)$

] odd

9. Determine if the following functions are odd, even or neither:

(a) $f(x) = \sin x \cdot (\sin x - 1)$

(b) $f(x) = e^{-x}(x-1)^{1/3}$

(c) $f(x) = \frac{2x}{|x|}$

(d) $f(x) = 2x^3 - 5x - 1$

(e) $f(x) = x(x-1)^{1/3}$

(b) Soln: $f(x) = x(x-1)^{1/3}$.

$f(-x) = (-x)(-x-1)^{1/3} \neq f(x) \neq -f(x)$

Neither even nor odd.

(d) Soln: $f(x) = 2x^3 - 5x - 1$

$f(-x) = 2(-x)^3 - 5(-x) - 1$

$= -2x^3 + 5x - 1 \neq f(x)$ [not even]

$= -(2x^3 - 5x + 1) \neq f(x)$ [not odd]

(c) Soln $f(x) = \frac{2x}{|x|}$

$f(-x) = \frac{2(-x)}{|-x|} = \frac{-2x}{|x|} = -\frac{2x}{|x|} = -f(x)$

This is odd function as $f(-x) = -f(x)$

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(b) $f(x) = e^{-x}(x-1)^{1/3}$

$f(-x) = e^{-(-x)}(-x-1)^{1/3} = e^{x}(-x-1)^{1/3} \neq f(x) \neq -f(x)$

$f(x)$ is neither even nor odd

(c) $f(x) = \sin(x)(\sin x - 1)$

$f(-x) = \sin(-x)\{\sin(-x) - 1\}$

$= -\sin(x)(-\sin x - 1) \neq -f(x)$ [not odd]

$= \sin x(\sin x + 1) \neq f(x)$ [not even]

* Definition of Fourier Series:

Let, $f(x)$ be defined in the interval $(-L, L)$ and determined outside of this interval by $f(x+2L) = f(x)$ i.e., assume that $f(x)$ has period $2L$. The fourier series or Fourier expansion corresponding to $f(x)$ is defined to be:

$$\frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right) \quad \text{---(1)}$$

Where the Fourier coefficients a_n and b_n are

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx \rightarrow n = 0, 1, 2, \ldots$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx \rightarrow n = 1, 2, 3, 4, \ldots$$

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